# MD SAJJAD HOSSAIN

**Data & Machine Learning | NLP | Human-in-the-Loop Annotation**

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LinkedIn | GitHub

## PROFESSIONAL SUMMARY

M.S. Computer Science candidate (expected Dec 2026, GPA 4.00/4.00) specializing in Applied NLP, Machine Learning, and Human-in-the-Loop systems. Proposed a novel annotation methodology for highly confidential, high-stakes data to systematically resolve ground-truth discrepancies. Engineered and deployed an automated crash analysis prototype for Austin TPW, streamlining their unstructured data pipelines. Proven ability to translate complex research into deployable, user-facing AI solutions through end-to-end ML development and full-stack prototyping.

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript, HTML, MATLAB
**ML & Deep Learning Frameworks:** PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face Transformers
**NLP & AI Expertise:** Multi-label Text Classification, Small Language Models (SML), TextCNN, FastText, Human-in-the-Loop Pipelines
**Data Science & Visualization:** Pandas, NumPy, Power BI, Matplotlib, Seaborn, Minitab
**Developer Tools & Prototyping:** Postman (API Testing), Figma (UI/UX), Git/GitHub, LaTeX

## EXPERIENCE

**Graduate Research Assistant** — **Aug 2024 – Present**
*Prairie View A&M University* — *Prairie View, TX*

- Analyzed and audited 10,000 police crash narratives, consolidating 72 original contributing-factor codes into an 8-class taxonomy and supporting a three-layer human-in-the-loop framework using the custom Linguistic Annotation for Classification and Entities (LACE) tool.
- Developed robust NLP pipelines utilizing FastText embeddings and a TextCNN classification model, achieving ~72% accuracy (FLAIRS-39) by proving direct multi-label classification outperformed hierarchical architectures on this specific dataset.
- Performed annotation-quality and discrepancy analysis that achieved near-perfect post-adjudication agreement (Cohen's $\kappa = 0.990$) and identified uncertainty as a strong signal for targeted expert review.
- Engineered and deployed a prototype crash-analysis system integrating Small Language Models (SML), Figma for UI/UX, and Postman for backend API testing, demonstrating full-pipeline execution from data processing to deployment [Watch Video Demo].

**Product Management Executive** — **Mar 2024 – Jun 2024**
*Barikoi Technologies Limited* — *Dhaka, Bangladesh*

- Collaborated with product and technical teams to translate product needs and feedback into improvement priorities, supporting iterative refinement of technology products.
- Worked across business and engineering stakeholders, strengthening requirements communication, issue tracking, and product-focused problem solving.

**Research Assistant** — **Dec 2021 – Feb 2022**
*Ahsanullah University of Science & Technology* — *Dhaka, Bangladesh*

- Supported experimental research, data analysis, laboratory operations, and student mentoring for predictive modeling and engineering optimization work.
- Applied response surface methodology, artificial neural networks, genetic algorithms, and particle swarm optimization to manufacturing-process prediction and optimization research.

## SELECTED TECHNICAL PROJECTS

**Skin Lesion Classification with Concept Bottleneck Models**
*Python, TensorFlow/Keras, EfficientNet-B4, CBM*

- Built a dermoscopic image classification pipeline integrating HAM10000 and ISIC datasets, utilizing Concept Bottleneck Models (CBM) for enhanced interpretability in medical diagnostics.
- Implemented transfer learning and fine-tuning with EfficientNet-B4 to deliver a strong balance of malignant-lesion detection and overall classification performance.

## EDUCATION

**Master of Science in Computer Science** — **Expected Dec 2026**
Prairie View A&M University, Prairie View, TX — GPA: 4.00/4.00

**Bachelor of Science in Industrial & Production Engineering** — **2022**
Ahsanullah University of Science & Technology, Dhaka, Bangladesh
GPA: 3.31/4.00 (ECE evaluated) | Relevant foundation: statistics, numerical analysis, operations research, supply chain management

## SELECTED PUBLICATIONS & CONFERENCE ACTIVITY

- **Hossain, M. S.**, Li, L., Perkins, J. A., Clary, J., & Meyer, J. (2026). "When Ground Truth Disagrees: A Human-in-the-Loop Audit of Annotation Errors in High-Stakes Crash Narratives." LAW XX, ACL, pp. 241–256. DOI
- **Hossain, M. S.**, Li, L., Perkins, J. A., Clary, J., Meyer, J., & Gomez, R. (2026). "Domain Adapted NLP for Multi-Label Crash Narrative Classification under Extreme Class Imbalance." FLAIRS-39. DOI
- **Hossain, M. S.**, & Hossain, N. U. I. (2024). "Intelligent Assessment Model to Investigate Mental Health: A PLS-SEM & ML Hybrid Approach." ASEM 2024.
- **Author Attendee, ACL 2026** – 64th Annual Meeting of the Association for Computational Linguistics, San Diego, CA, Jul 2026; presented first-author LAW XX work in the workshop program.
- **Author Attendee, FLAIRS-39** – 39th International Florida Artificial Intelligence Research Society Conference, 2026; presented first-author work on domain-adapted NLP and multi-label classification.

**U.S. Work Authorization:** Available to work pursuant to valid CPT/OPT authorization; employer visa sponsorship is not required during the authorized CPT/OPT period.